# PRACTICE ASSIGNMENT - SAMPLE ONLY

## TMA Workspace | Fictional Mathematics Assignment

This document is entirely fictional and is not an Open University assignment. It is intended for testing PDF display, navigation and answer entry.

## Instructions

Answer all questions. Show your working where appropriate. Use the separate answer editor in TMA Workspace to record your responses.

## Question 1 - Fractions and percentages (20 marks)

(a) Simplify the fraction 18/24. (4 marks)

(b) Calculate 35% of 240. (4 marks)

(c) A notebook costs GBP 12.80. Its price increases by 15%. Calculate the new price. (6 marks)

(d) Explain the difference between increasing a quantity by 20% and finding 20% of that quantity. (6 marks)

## Question 2 - Geometry (20 marks)

(a) A rectangle is 8.5 cm long and 4 cm wide. Find its area. (5 marks)

(b) A circle has a diameter of 6 cm. Calculate its circumference to two decimal places. (5 marks)

(c) A cylinder has radius 3 cm and height 10 cm. Calculate its volume in cubic centimetres. (6 marks)

(d) State the units you would use for area and for volume, and explain why they differ. (4 marks)

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## Question 3 - Interpreting data (30 marks)

A student records the number of minutes spent studying on five days:

| Day | Monday | Tuesday | Wednesday | Thursday | Friday |
|---|---|---|---|---|---|
| Minutes | 40 | 55 | 35 | 70 | 50 |

(a) Calculate the total study time over the five days. (6 marks)

(b) Calculate the mean daily study time. (6 marks)

(c) Identify the day with the longest study time. (4 marks)

(d) Describe one limitation of using only five days of data to estimate a typical week. (8 marks)

(e) Suggest a suitable chart for displaying these data and explain your choice. (6 marks)

## Question 4 - Algebra (30 marks)

(a) Simplify 3x + 5x - 2. (5 marks)

(b) Solve 4x + 7 = 31. (7 marks)

(c) Evaluate 2a squared + 3 when a = 4. (8 marks)

(d) A taxi charges GBP 3 plus GBP 1.80 per kilometre. Write an expression for the total cost of a journey of d kilometres, and find the cost of a 12 km journey. (10 marks)

End of fictional practice assignment.